BIOINFORMATICS PROJECT REPORT

SeqLoom

NGS Sample Quality Monitoring

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PROJECT SUMMARY

SeqLoom is a SQL-driven NGS sample quality monitoring and assessment prototype. It organizes selected FastQC, trimming and alignment indicators in SQL Server, applies transparent provisional review rules, and presents the results through an interactive Power BI dashboard.

8	7	87.43%
Samples represented	Samples marked for review	Average unique mapping*

*Values reflect the previously recorded SQL assessment view and should be reconciled with the final Power BI screenshot before public release.

Prepared by

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Scope
A learning and portfolio prototype for organizing and reviewing NGS sample quality indicators. It is not a clinical diagnostic system and does not replace laboratory quality procedures or expert interpretation.

Project technologies: SQL Server · T-SQL · Power BI · DAX · FastQC · NGS QC concepts

1. Problem statement and project objectives

Problem. Next-Generation Sequencing (NGS) workflows produce multiple quality indicators at different stages. When these indicators remain in separate outputs or tables, comparing samples and identifying which ones need closer inspection can become time-consuming. SeqLoom brings selected indicators into a structured data model and a single interactive report.

Why this matters

- A reviewer needs to compare samples consistently rather than inspect disconnected metrics one at a time.
- A category-level summary is useful, but the reviewer must also be able to inspect the sample-level evidence behind it.
- Assessment logic should be visible and explainable. A dashboard category should not appear as an unexplained or definitive verdict.

Project objectives

- Organize selected FastQC, trimming and alignment metrics in SQL Server tables.
- Combine relevant fields into a reusable sample-level SQL view.
- Summarize sample counts by provisional QC assessment category.
- Compare unique mapping percentages and inspect individual sample details in Power BI.
- Demonstrate assessment snapshot and rule-version tracking for traceability.

2. Data and database structure

The project database is SeqLoomDB. Four source tables hold the selected metric groups. SQL views prepare the assessment outputs consumed by Power BI, while two audit-oriented tables hold run metadata and sample assessment history.

Database object	Role in SeqLoom
dbo.fastqc_modules	Stores module-level FastQC results.
dbo.fastqc_basic_metrics	Stores basic FastQC metrics.
dbo.trimming_metrics	Stores trimming-related metrics.
dbo.alignment_metrics	Stores alignment and mapping statistics.
dbo.vw_SampleQC_Assessment	Combines selected QC and alignment indicators and assigns a provisional sample assessment.
dbo.vw_QC_Assessment_Summary	Aggregates sample counts by assessment category.
dbo.QC_Assessment_Runs	Records assessment-run metadata, including timestamp and rule version.
dbo.QC_Assessment_History	Stores sample-level assessment history.

The first recorded assessment snapshot used RunID 1 and rule label v1-provisional (recorded on 23 October 2026). This demonstrates a traceability concept; it should not be described as a fully automated production audit platform unless the full process is tested end to end.

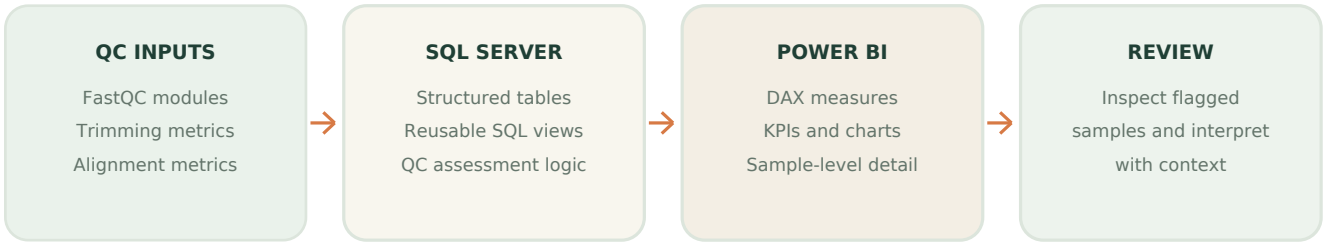
3. Methodology

SeqLoom follows a data-integration and reporting workflow. The project begins with prepared sample-level QC metric outputs, stores them in relational tables, combines the metrics using SQL views, applies provisional review logic, and displays the resulting overview and details in Power BI.

Method steps

Step	Activity	Purpose
1. Metric collection	Collect selected FastQC, trimming and alignment metrics for each sample.	Bring relevant QC evidence into one project dataset.
2. SQL storage	Store the metric groups in separate SQL Server tables.	Keep the data structured and reusable.
3. SQL integration	Use vw_SampleQC_Assessment to combine sample-level fields.	Create a consistent sample-level assessment dataset.
4. Summary logic	Use vw_QC_Assessment_Summary to count samples by category.	Provide an overall view of the current assessment results.
5. Snapshot tracking	Record assessment-run metadata and a rule-version label.	Make the assessment context more traceable.
6. Power BI reporting	Load the SQL views and create DAX measures, charts, KPIs and a sample selector.	Make the results easier to explore and communicate.
7. Review	Filter to a sample and inspect its underlying QC indicators.	Support follow-up and interpretation by a reviewer.

End-to-end workflow



Traceability concept: assessment run + timestamp + rule-version label

Key Power BI measures

Samples Requiring Review counts distinct samples classified as Needs Investigation or Review Required. Average Unique Mapping calculates the average unique mapping percentage in the current filter context. These measures respond to report filters when the model relationships and visual interactions are configured correctly.

Interpretation note: the metric names and calculations describe the project report; they do not validate the thresholds used to assign the provisional categories.

4. Results and dashboard outputs

The previously recorded SQL assessment summary contains eight samples. Five were assigned Needs Investigation, two Review Required and one Initial Review Passed. The dashboard was designed to show these category totals alongside sample-level evidence and mapping comparisons.

Assessment category	Samples	Share of total
Needs Investigation	**5**	62.5%
Review Required	**2**	25.0%
Initial Review Passed	**1**	12.5%
Total	**8**	**100%**

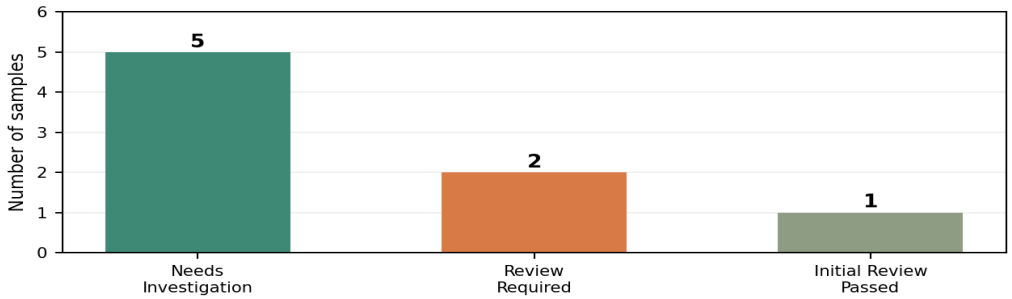


Figure 1. Previously recorded SQL assessment category counts. Categories are provisional triage labels.

Sample-level unique mapping results

The earlier SQL view recorded the following unique mapping percentages. They provide a comparison across the eight sample IDs and are not, by themselves, a pass/fail criterion.

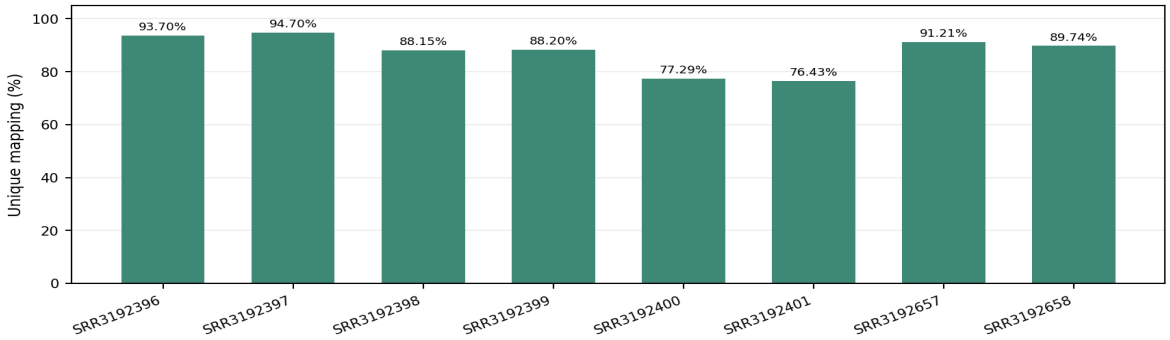


Figure 2. Unique mapping percentages from the previously recorded SQL view.

Sample ID	Unique mapping	Sample ID	Unique mapping
SRR3192396	93.70%	SRR3192400	77.29%
SRR3192397	94.70%	SRR3192401	76.43%
SRR3192398	88.15%	SRR3192657	91.21%
SRR3192399	88.20%	SRR3192658	89.74%

KPI summary: 8 total samples; 7 require review; average unique mapping is about 87.43%. Review combines Needs Investigation and Review Required.

5. Dashboard interpretation

The Power BI report combines a high-level view with sample-level detail. Each visual has a specific role:

Dashboard element	What it communicates
Total Samples KPI	Number of distinct samples represented in the current report context (expected: 8).
Samples Requiring Review KPI	Distinct samples classified as Needs Investigation or Review Required (previous summary: 7).
Average Unique Mapping KPI	Average of the unique mapping percentage in the current filter context (previous result: about 87.43%).
Assessment overview chart	Displays the 5 / 2 / 1 provisional category distribution.
Unique mapping chart	Compares unique mapping percentages between samples.
Sample QC details table	Shows the sample assessment and supporting indicators such as mapping, unmapped-too-short percentage and trimmed fail count.
Sample ID selector	Filters report visuals to an individual sample when interactions are enabled.

6. Limitations and responsible interpretation

- Provisional rules: The assessment thresholds are project rules and have not been clinically or externally validated.
- QC flags require context: A FastQC FAIL/WARN flag or a mapping percentage alone cannot determine whether a sample is usable.
- Mapping is context-dependent: Results can be influenced by read quality, reference selection, contamination, library preparation, biological characteristics and pipeline settings.
- Prototype scope: SeqLoom is a learning and portfolio prototype, not a validated laboratory information system or clinical decision-support product.
- Audit maturity: Run and history tables demonstrate a traceability design; production automation and operational controls require further testing.

Before public release: data consistency check

A previous dashboard screenshot appeared to show sample IDs that differ from the earlier SQL view results. Before publishing, reconcile the dashboard, charts and table against the current SQL views. Confirm that the same eight IDs appear in every visual, that category counts sum to eight, and that percentage values are not being summed across samples.

Release check	Expected verification
Sample IDs	Confirm the eight IDs shown in the report match the SQL source.
Category totals	Confirm 5 + 2 + 1 = 8 for the same data snapshot.
Percentages	Display per-sample mapping values; do not add percentages across samples.
Filter behavior	Check that selecting a sample filters the intended visuals and table rows.

7. Conclusion and next steps

SeqLoom demonstrates how selected NGS quality indicators can be organized in SQL Server, combined into reusable views, summarized using transparent provisional rules, and communicated through an interactive Power BI dashboard. The project connects bioinformatics workflow knowledge with practical skills in relational data organization, SQL querying, DAX calculations and visual analytics.

The key outcome is a single reporting workflow that helps a reviewer move from a sample-level overview to the metrics behind an individual assessment. Its value is the integration of bioinformatics context, explainable rules, interactive visual reporting and an initial assessment-history concept - not a claim that the tool makes clinical decisions.

Recommended future improvements

- Validate and document assessment rules with qualified domain experts before treating any threshold as a formal QC criterion.
- Add automated checks for missing, duplicated or inconsistent sample records and data freshness.
- Test the SQL-to-Power BI refresh and audit snapshot process as a repeatable end-to-end workflow.
- Expand the project with additional sample sets and document pipeline versions and reference context.
- Improve dashboard labels, percentage formatting and drill-through so that each displayed value can be traced to its source metric.

Conclusion
SeqLoom turns scattered NGS quality indicators into a structured, reviewable and easy-to-explore report. It is a practical bioinformatics portfolio project that demonstrates data handling, SQL, Power BI and careful interpretation of quality metrics.

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Report note: This document summarizes the project structure and previously recorded SQL outputs available at the time of preparation. Verify current source data and dashboard consistency before using numerical outputs in a public portfolio or presentation.